


ETHEREUM ECOSYSTEM GRANT APPLICATION

# Ethereum-native civic infrastructure

21 lets anyone report a real-world problem, fund the fix, hire the people who do the work, and verify the result — settled on Ethereum.

| ASK | STAGE | LICENSE |
| --- | --- | --- |
| $75,000 / 10 mo. | MVP + 3 city pilots | Open source · MIT |

01 — THE PROBLEM

# Cities can see the problem. They cannot coordinate the fix.

01 **Reporting ends in a ticket.**

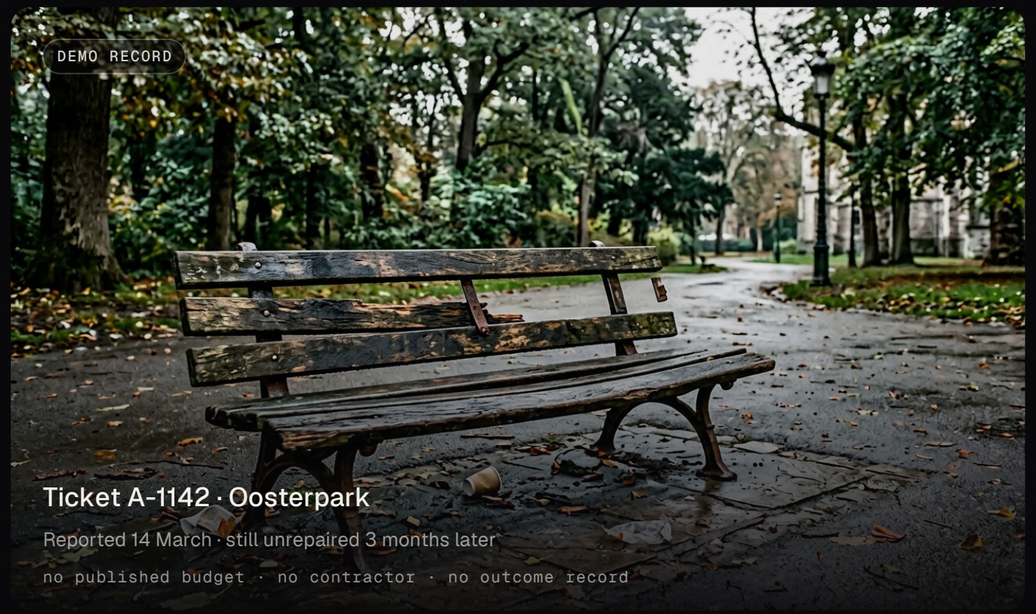
A photo to a hotline produces a reference number, not a budget. Residents never learn who decided, who was paid, or what actually happened.

02 **Small works fall through.**

A €420 bench, a €950 bike rack, a €300 mural: too small for procurement, too visible to ignore. They wait for a season that never comes.

03 **Public-goods funding stops at the screen.**

Crypto has capital, matching pools and governance tooling for digital work. There is no delivery layer for physical work.




WHY THIS IS BUILDABLE NOW

- **L2 fees** — a €4 settlement is economically sane
- **Personhood** — proof-of-personhood is usable
- **Attestations** — EAS makes verified claims portable
- **Funding** — QF + retro funding is mature

# A public protocol for getting real things done.

21 turns a geotagged report into a funded, contracted and verified piece of work. The money moves through an on-chain escrow; the decision, the evidence and the payout stay public.

01 **Report**
Photo, location, category, urgency

02 **Propose**
Contractors publish budget + milestones

03 **Fund**
Community + matching pool fill escrow

04 **Decide**
Eligible participants select a proposal

05 **Build**
Work happens in the physical world

06 **Verify**
Evidence + quorum of attestations

07 **Settle**
Tranche released to the contractor

### Non-custodial
21 never holds funds. A per-task escrow contract does.

### Rule-bound
Verification and payment rules are pinned before funding opens.

### Composable
Any city, DAO or funder can plug into the same registry.

### Forkable
MIT-licensed contracts and client. No lock-in for municipalities.

**SAME REGISTRY, MANY FRONT DOORS**

A city can run its own branded instance, a funder can open a round, a contractor can use a mobile-only client — all writing to one task registry.

03 — ONE TASK, END TO END

# A broken bench in nine days.

Demo scenario built from the product spec. Amounts and dates are illustrative, not reported results.




Before — three missing slats, reported from the phone




After — replaced and re-fixed by Groenwerk BV

TASK LEDGER · PUBLIC BY DEFAULT

| | | |
|---|---|---|
| Day 0 | Task published | €420 target |
| Day 1 | 3 proposals received | €380 - €455 |
| Day 3 | 84 residents voted | 62% Groenwerk |
| Day 4 | Escrow funded | €420 in USDC |
| Day 7 | Milestone evidence | 6 photos + GPS |
| Day 9 | Dual attestation | payment released |

WHAT THE RESIDENT GETS

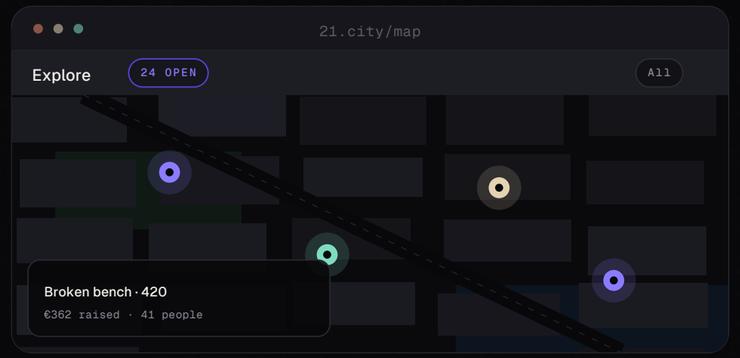
A public URL with the full history, the contractor's identity and the final payment.

WHAT THE CONTRACTOR GETS

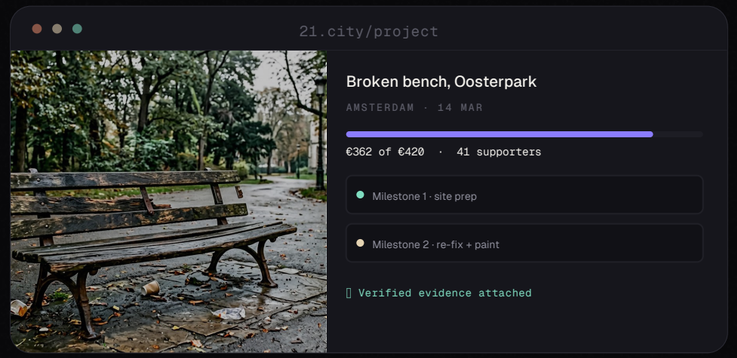
Milestone payments released in stablecoin the day verification clears — no 90-day invoice cycle.

# Six surfaces, one continuous workflow.

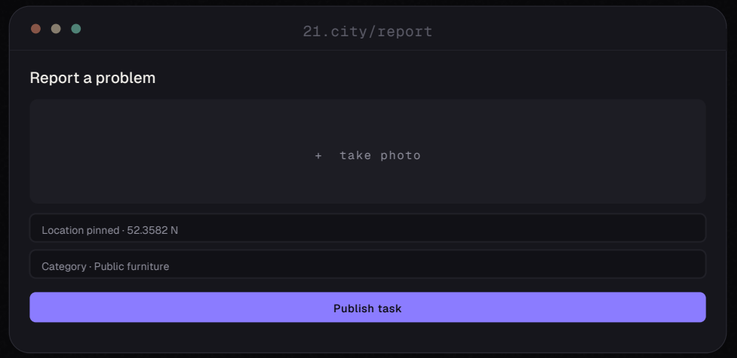
Responsive web app plus an installable mobile PWA. Every screen below is part of the MVP scope, not a future release.



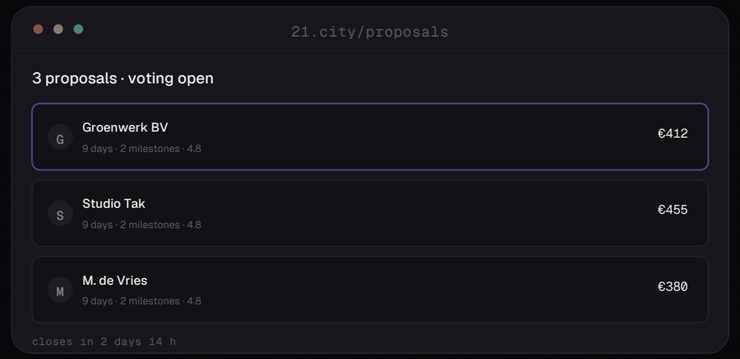

01 **Explore** Map-first discovery with live task markers



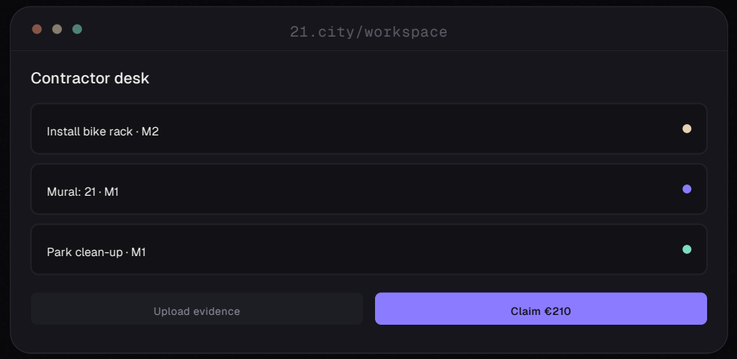

02 **Project** Evidence, budget, milestones, discussion



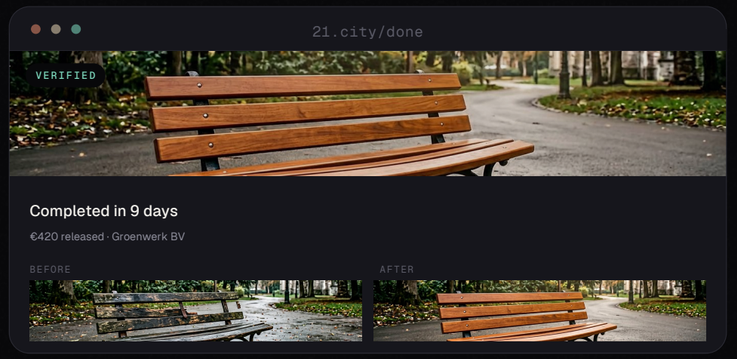

03 **Report** Photo, GPS, category — under 60 seconds




04 **Proposals** Competing bids with itemised budgets




05 **Contractor** Bids, contracts, evidence, payouts




06 **Completed** Before / after and the payment record

# Money moves only when the street agrees.

A photo alone is not proof. 21 combines device-signed capture, geofencing, independent attestors and an economic challenge window.

**01 Device-signed capture**

Photos are taken in-app: EXIF, GPS, gyroscope and a hardware-backed signature are hashed and stored with the file.

**02 Proximity, not claims**

The capture must fall inside the task geofence; high-value tasks require a reviewer visit.

**03 Dual attestation**

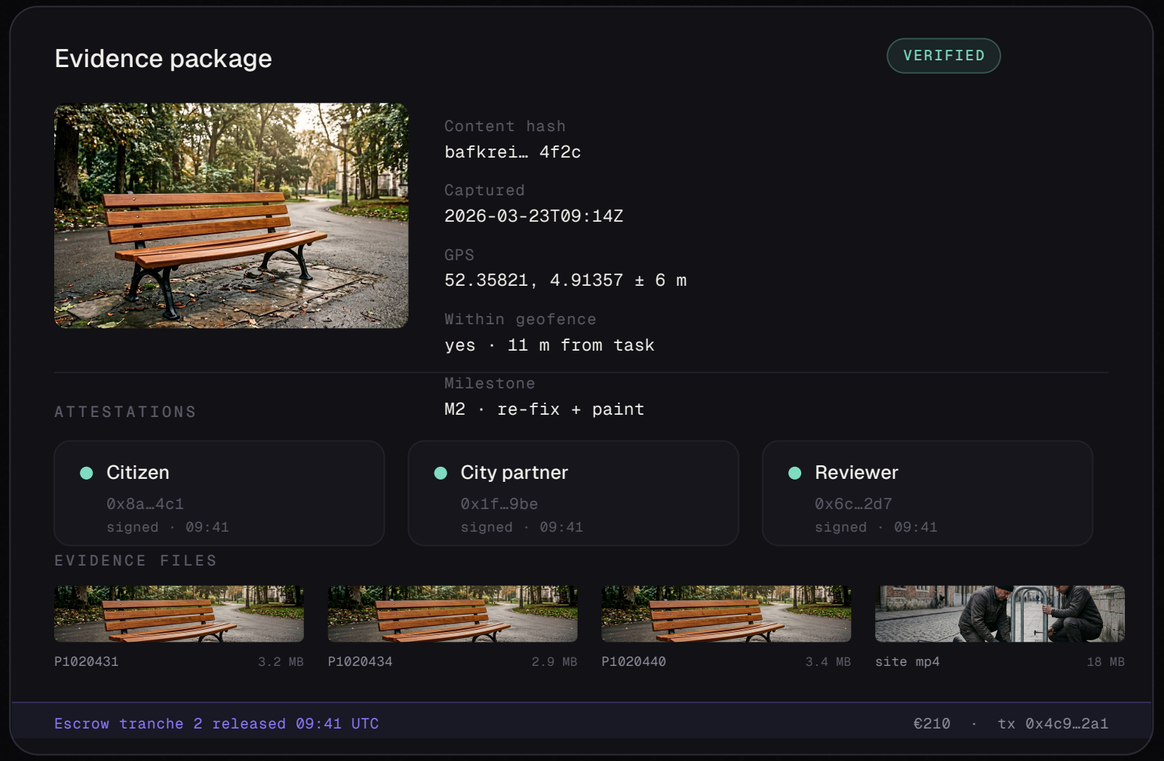
A citizen attestor and a city partner sign the milestone. A random reviewer is added above €1,000.

**04 Challenge window**

72 hours to dispute with a staked bond. Losing a dispute costs the bond; winning returns it plus a share of the fee.

**05 Reputation that cannot be bought**

Scores come from verified work only (ERC-5192, non-transferable) and decay without activity.



# Funding buys outcomes, not votes.

Each project publishes its rules before the first euro arrives. The rules hash is pinned on-chain, so the game cannot change mid-round.

01

## Who may vote

Three configurable profiles

- Resident: proof-of-personhood inside a 2 km radius
- Contributor: any wallet above a €5 floor
- Partner: city wallet, advisory weight only

02

## How money is weighted

Default profile: one person, one vote

- A €5 and a €500 contribution carry equal voice
- Token-weighted modes require an explicit opt-in by the city
- Quadratic matching from the 21 pool, capped per project

03

## How a winner is chosen

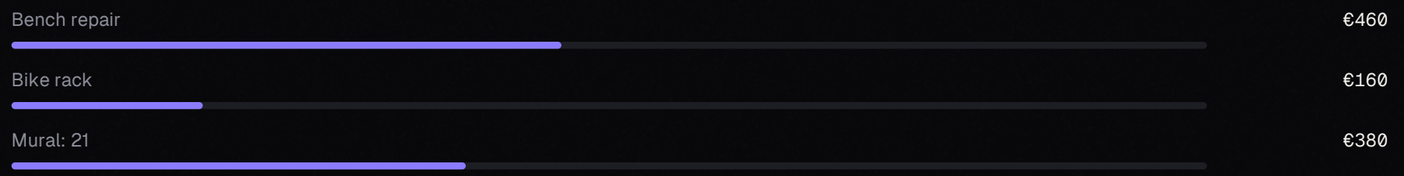
Approval vote, 5-day window

- Quorum: 25% of eligible voters, minimum 20 people
- No quorum → partner + council decide, rationale published
- Losers' proposals stay open for other rounds

### QUADRATIC MATCHING, ILLUSTRATED

Pool €1,000 · 3 projects

Every contribution is square-rooted before it counts, so twenty small donors outrank one large one.




RULES HASH `0x9d2e71…c40b` · pinned at funding open · quorum 25% · 5-day vote · one-person-one-vote — editable only with 72h timelock

# 21 is a number anyone can paint.

A commissioned, always-authorised mural programme runs alongside the protocol. It is how a civic tool becomes recognisable on a street in Lagos or Lisbon without a translation.

01 **Commission, never tag**

Every wall has a signed owner consent, a removal clause and a paid artist. The registry refuses locations without a permission record.

02 **21 murals in 21 cities**

Year one: €300–€1,200 per wall, funded by the campaign pool and local sponsors. Each is a Proof-of-Place entry on-chain.

03 **The number is the message**

No slogan to translate, no logo to explain. Two digits, painted by hand, pointing back to a working protocol.




| 21 | €300–1,200 | 100% | 1 |
|---|---|---|---|
| cities in year one | artist fee per wall | consented walls | permanent registry entry per mural |

# Serious enough for a treasury. Simple enough for a phone.

A compact system ships with the MVP: tokens, components, interaction states and usage rules, documented in a public Storybook.

WORDMARK

21

Build the world together.

Lockup: app icon · sticker · wall · favicon

Geist Sans + Geist Mono · OFL · no gradients, no glow

PALETTE

- Paper #F3F1EC
- Graphite #16161C
- Signal violet #8B7CFF
- Deep violet #5343D6
- Verified #7FDCC0
- Sand #E4D3B2

TYPE SCALE

Display 56 / 60

Title 30 / 36

Body 13 / 19

LABEL 8.5 / MONO / 1.5

COMPONENTS

REPORT AN ISSUE | EXPLORE MAP | VERIFIED | VOTING OPEN | PENDING · 12s | INSUFFICIENT FUNDS

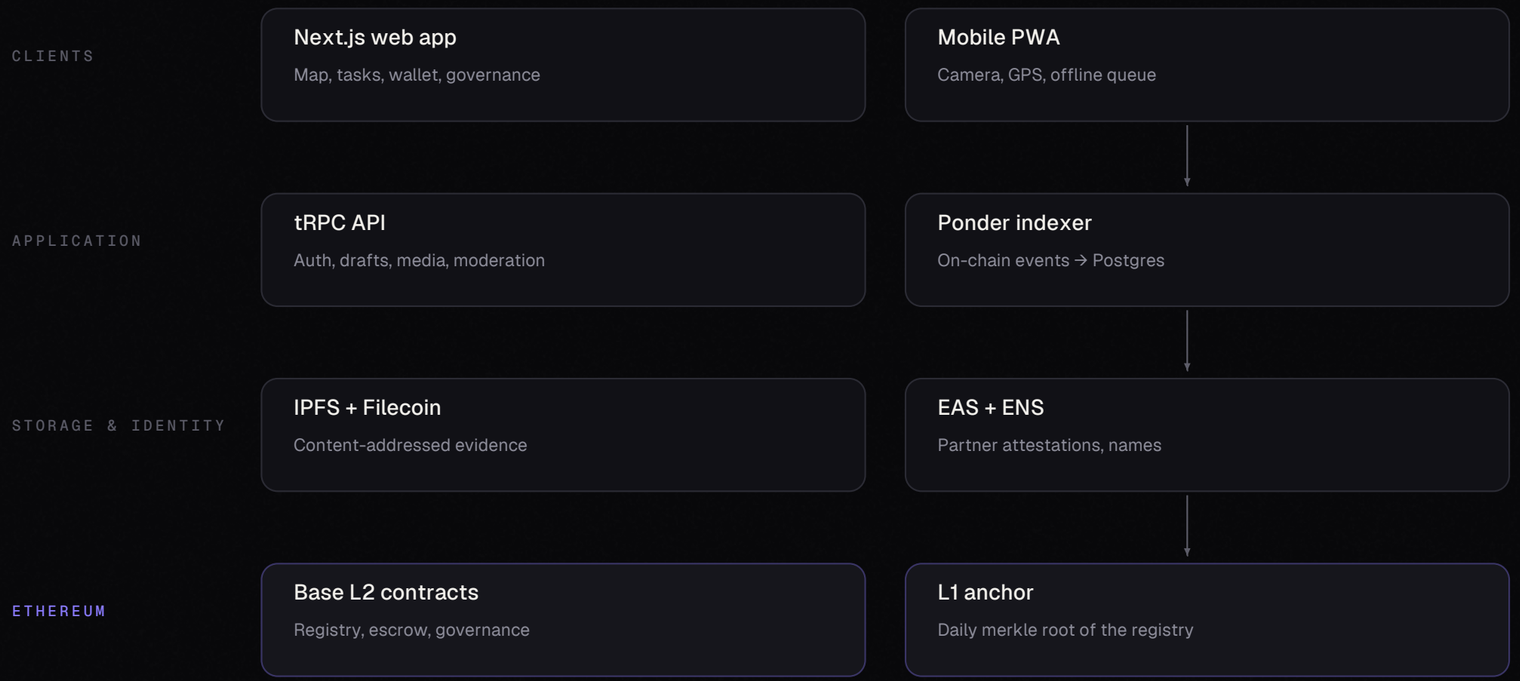
LIFECYCLE

draft → proposals → voting → funding → in progress → verification → settled

one colour, one word, one next action per state

# Boring where it matters. Verifiable where it counts.

TypeScript end to end, Foundry for contracts, no custodial backend.




CONSTRAINTS

Gas per task
< $0.02 on Base

Contract standard
Solidity 0.8 / Foundry

Upgrade policy
Immutable core + factory

Data on-chain
State + hashes, never media

Who can pause
3-of-5 guardian, 48h delay

TRUST BOUNDARY

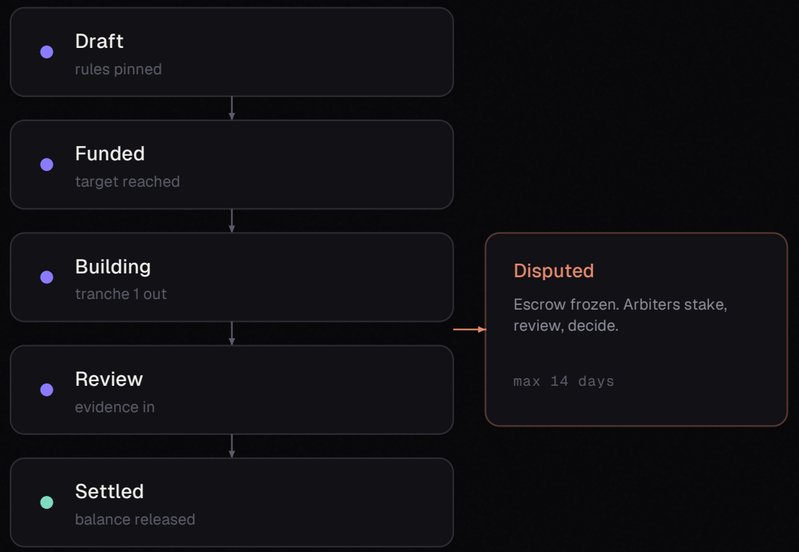
Funds live in escrow contracts. The 21 server can censor a list, but it cannot move, freeze or redirect a single euro.

# Eight small contracts, one escrow per task.

Immutable core deployed by a factory, so each task gets its own escrow address and its own accounting. No proxy admin can touch user funds.

| | |
|---|---|
| `TaskRegistry` | Canonical task record: geohash, category, media CID, status |
| `ProposalBook` | Bids, itemised budgets, milestone schedule, withdrawal |
| `FundingPool` | Contributions, refunds on failure, matching accounting |
| `Escrow` | Per-task USDC, milestone tranches, release conditions |
| `Verification` | Attestation quorum, evidence CIDs, challenge window |
| `Dispute` | Staked arbiters, slashing, escalation to council |
| `Reputation` | Non-transferable score from verified completions |
| `Treasury` | Matching pool, Safe multisig + 72h timelock |

ESCROW LIFECYCLE




Escrow rules are hashed and emitted on TaskPublished. A task cannot be funded until its payment logic is immutable.

# The attacker is a person with a camera.

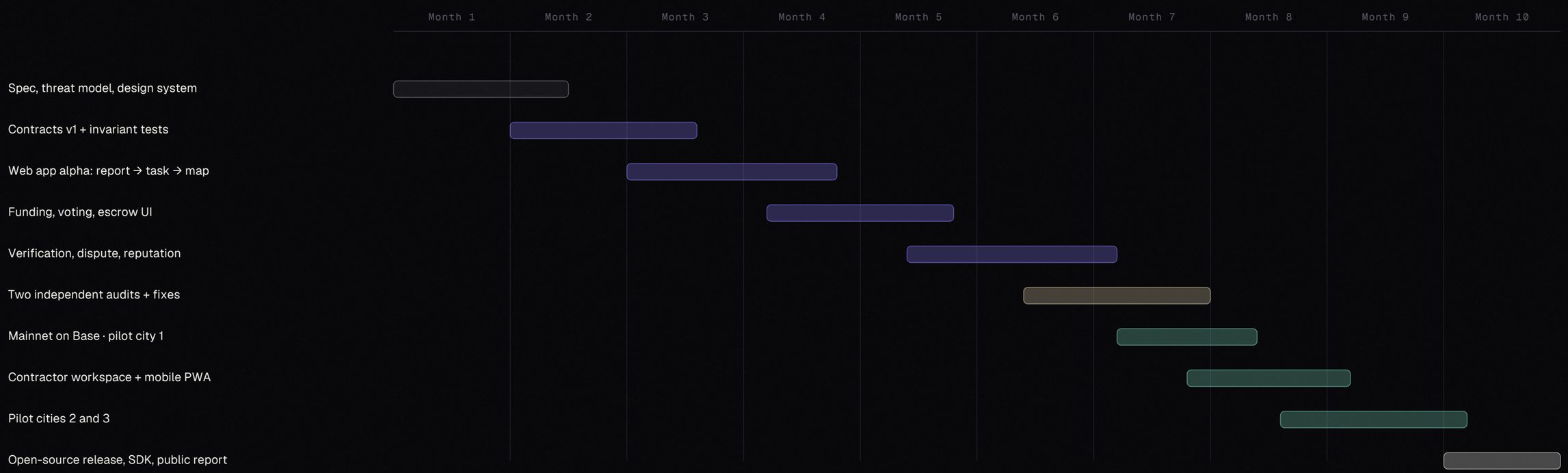
Physical-world verification has adversarial failure modes that a normal audit does not catch. Each one is handled explicitly.

| THREAT | WHY IT IS REAL | MITIGATION IN THE MVP |
|---|---|---|
| **Fabricated or AI-generated proof** | A model can draw a fixed bench in seconds. | In-app capture only, signed device metadata, C2PA content credentials, phash dedupe, human spot audits. |
| **GPS spoofing** | Mock locations are one developer setting away. | Geofence + cell/Wi-Fi corroboration, on-site reviewer above €1,000, anomalous-radius flags. |
| **Reporter–contractor collusion** | The easiest way to extract money from a pool. | Role separation, random reviewer panel, reputation at risk, payout caps for new addresses. |
| **Sybil voting** | Free wallets, cheap votes. | Personhood-gated eligibility, snapshot at vote open, one-person-one-vote as the default. |
| **Escrow bug or exploit** | Contracts hold real money on day one. | Two independent audits, invariant fuzzing, per-task caps, 3-of-5 guardian pause with 48h delay. |
| **Privacy of residents** | Photos contain faces, plates and windows. | On-device blur, coarse public coordinates, off-chain personal data, GDPR DPIA before pilot 1. |

Open question we bring to reviewers: what is the minimum viable proof for a €300 repair that still resists a determined faker?

# Ten months from spec to three live cities.

Every milestone ships something a reviewer can open, test or audit. Nothing is marked done without a public artefact.




RELEASE GATES

**Month 6** audits closed · no critical or high findings open
milestone M5

**Month 7** first real euro spent through escrow
milestone M6

**Month 10** three pilots, public report, open SDK
milestone M8

# What the grant actually pays for.

Acceptance criteria are written as artefacts, not intentions. Each milestone is reviewed before the next tranche is released.

| | MILESTONE | ACCEPTANCE CRITERION | SHARE |
|---|---|---|---|
| M1 | Protocol spec + threat model | Published RFC, 2 external reviews | 5% |
| M2 | Contracts on testnet | 90%+ coverage, invariants fuzzed, testnet deploy | 10% |
| M3 | Alpha app: report → task → escrow | A bench task completed end-to-end on testnet | 15% |
| M4 | Verification + dispute modules | Dual attestation and a resolved dispute in a live drill | 10% |
| M5 | Two independent audits | Reports published; all critical/high fixed | 20% |
| M6 | Base mainnet launch | Verified contracts, per-task cap, guardian live | 10% |
| M7 | Three city pilots | 3 partners, 20 tasks each, real payouts | 20% |
| M8 | Open source + SDK | MIT release, docs, 3 external integrators | 10% |

Cumulative release: 15% by M2, 40% by M4, 70% by M6, 100% at close-out. An unmet criterion means an unpaid milestone, not a renegotiation.

14 — BUDGET

# $75,000, allocated line by line.

People, audits and pilots. No token, no marketing spend, no margin hidden in the total.



Total $75,000 · 10 months · 2 FTE + 1 part-time

| LINE ITEM | SHARE | USD |
|---|---|---|
| Smart contracts, tests, two audits | 32% | $24,000 |
| Product, frontend, mobile PWA | 24% | $18,000 |
| Backend, indexer, storage pipeline | 13% | $9,500 |
| Verification, dispute and reviewer tooling | 10% | $7,500 |
| Three city pilots and the mural programme | 12% | $9,000 |
| Design system, documentation, Storybook | 5% | $4,000 |
| Legal, GDPR review, operations, travel | 4% | $3,000 |

If the grant is smaller than requested, the audits and the mainnet launch are the last things we cut.

15 — TEAM

# Three roles, named before the first tranche.

21 is a small team plus one municipal advisor. Replace the bracketed fields with names and profile links before submission.

P

### Protocol lead

contracts

[name] · [github / linkedin]

Solidity and Foundry, escrow design, one prior public-goods deployment. Owns M1–M6 and both audits.

commitment: full time, 10 months

P

### Product & design lead

product

[name] · [github / linkedin]

Civic interfaces, map and form UX, accessible design systems. Owns M3, M7 and the pilot onboarding.

commitment: full time, 10 months

F

### Full-stack engineer

backend

[name] · [github / linkedin]

Next.js, indexer, media pipeline, PWA capture and offline queue. Owns M3 and M7.

commitment: full time, 10 months

ADVISOR

**Municipal procurement + public-art permissions**

A former city procurement officer reviews pilot contracts, consent templates and the mural programme before any wall is touched. [name · organisation]

HOW WE WORK

- Public roadmap, public repo, weekly demo
- Every payment in milestone tranches
- No token, no presale, no paid listings

OPEN ROLE FUNDED BY TRANCHE 2

Verification & field-operations lead — runs reviewer panels in the three pilot cities.

NOT APPLYING FOR

Marketing budget, token liquidity, or exchange listings.

16 — SUCCESS METRICS

# Targets, not traction.

21 is pre-launch. These are the numbers we commit to reporting at close-out, with the methodology published alongside them.

**3**
city partners under a signed pilot agreement
source: signed pilot MOUs

**60**
tasks published, verified and settled
source: registry events

**€45k**
value moved through escrow
source: escrow contract balance

**25**
contractors paid, median payout < 24h
source: payout timestamps

**21**
commissioned murals registered on-chain
source: proof-of-place entries

**500**
unique participants with proof-of-personhood
source: eligible voter snapshot

GUARDRAIL METRICS — WE FAIL IF THESE MOVE THE WRONG WAY

| | | | |
|---|---|---|---|
| **< 1% of released value** | **< 8% of tasks** | **< 72h median** | **> 40%** |
| Fraud rate | Dispute rate | Time to first proposal | Repeat contractors |

# Because the alternative is a database someone else controls.

A civic platform that holds public money should not be able to rewrite its own history. Ethereum gives 21 four properties no SaaS can offer.

### 01 Neutral settlement

No company sits between a resident's euro and a contractor's invoice. The escrow is the counterparty.

### 02 A global payment rail

USDC on Base pays a contractor in Nairobi or Nijmegen in seconds, without a bank account.

### 03 Composable public goods

Quadratic funding, retro funding, EAS attestations and Safe treasuries plug in unchanged.

### 04 Permanent public record

What was promised, funded and delivered stays queryable — even if 21 disappears.

**WHAT 21 DELIBERATELY DOES NOT DO**

- × No token sale and no speculative asset
- × No custody of user funds at any point
- × No paid placement of projects or contractors
- × No growth metrics that are not on-chain verifiable

**WHERE THE GRANT FITS**

- 21 is public-goods infrastructure: an open registry and escrow any city, DAO or funding round can build on.
- Public-goods rounds can fund tasks directly through the same escrow — no new rails, no new audit surface.
- The grant funds the protocol and the first three proofs, not a company's growth.

# Built to outlive the funding round.

The protocol is open source and unowned. Revenue, if any, is optional and capped; the public instance stays free for residents.

01

### Sustainable by design

- Optional 1.5% fee on released escrow, capped and opt-in per city
- Hosted instance + support subscription for municipalities
- Matching pools funded by partners, not by the protocol

02

### Progressive decentralisation

- M1–M6: core team ships, guardian multisig holds pause rights
- M7–M10: council elected by contributors, parameters on-chain
- Year 2: registry ownership renounced; fee switch under token-less vote

03

### What stays open

- Contracts, indexer and client under MIT
- Public data export: every task, attestation and payment
- A documented schema so other cities can fork without us

IF THE PILOT WORKS, THE NEXT EIGHTEEN MONTHS LOOK LIKE

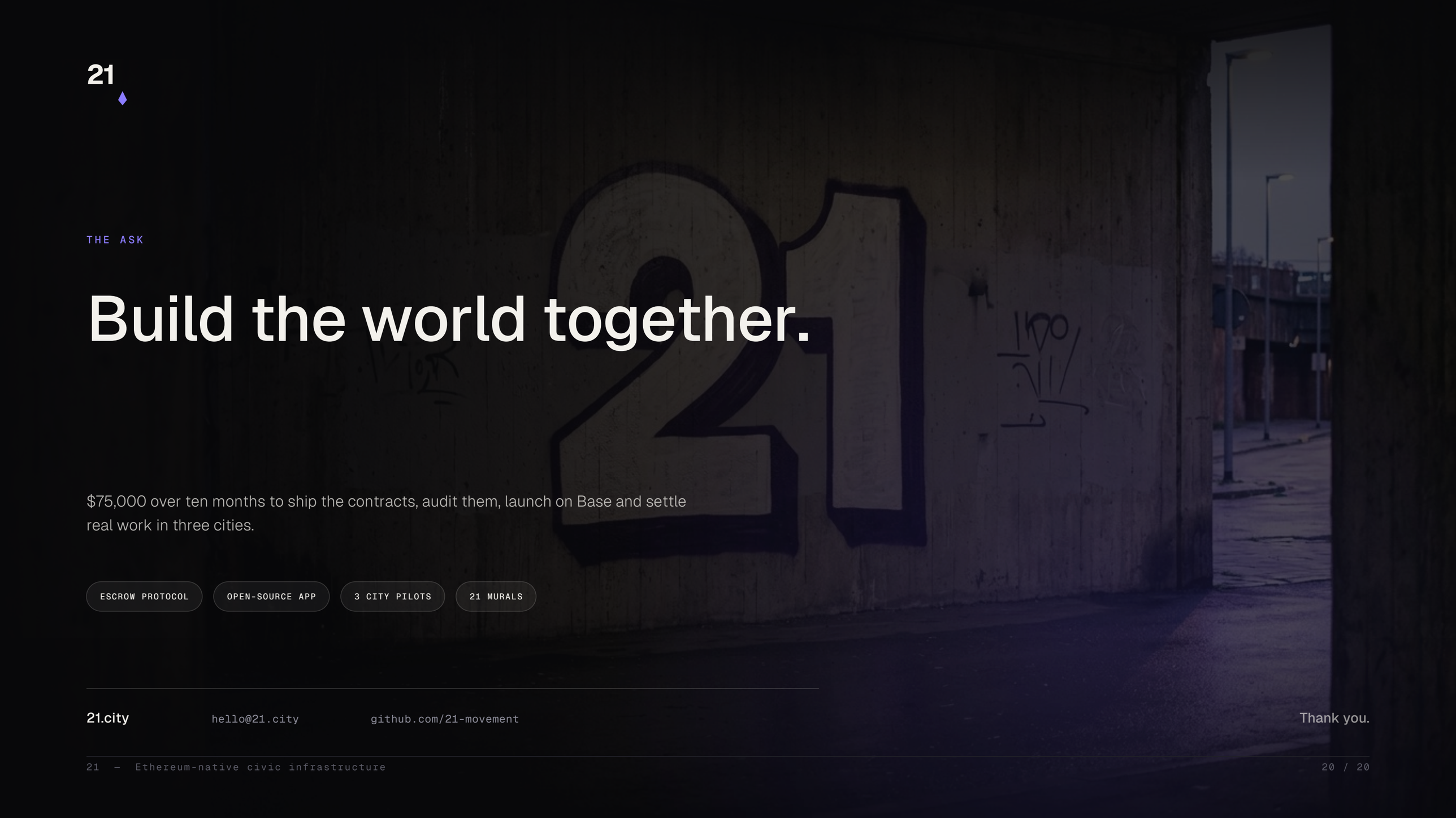
**10 cities**
EU + one emerging-market municipality

**300 tasks**
recurring maintenance, not one-off demos

**€1m escrow**
mostly partner-funded matching pools

**1 SDK**
any civic org can run their own instance

The registry stays public, the code stays MIT, and the three pilots decide whether this is real.



21

THE ASK

# Build the world together.

$75,000 over ten months to ship the contracts, audit them, launch on Base and settle real work in three cities.

ESCROW PROTOCOL · OPEN-SOURCE APP · 3 CITY PILOTS · 21 MURALS

21.city hello@21.city github.com/21-movement

Thank you.